

ISHANK MISHRA

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PROFESSIONAL SUMMARY

AI/ML Engineer and Full Stack Developer specializing in LLM orchestration, multi-agent systems, and end-to-end AI application development. Six months of hands-on internship experience applying computer vision and deep learning in a production setting, alongside independent work building production-grade RAG pipelines, fine-tuned LLMs, and AI-powered web applications using FastAPI, React, and modern Python stacks. Passionate about agentic AI, backend architecture, and deploying scalable intelligent systems that solve real-world problems.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

AI / LLM: LangChain, LangGraph, CrewAI, AutoGen, OpenAI APIs, Gemini API, Ollama, RAG, QLoRA Fine-Tuning, PEFT, Prompt Engineering

ML / DL: TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV • CNNs, RNNs, Transformers, Transfer Learning, Multi-Agent Systems

Full Stack: FastAPI, React, Next.js, Vite, Tailwind CSS, SQLAlchemy, REST APIs, Streamlit

Databases: PostgreSQL, SQLite, FAISS, ChromaDB, Pinecone

DevOps / Tools: Docker, Git, GitHub, GitHub Actions, AWS (EC2, S3, Lambda), VS Code, Jupyter

EXPERIENCE

AI/ML Intern — Labmentix

1 June 2026 – 1 Aug 2026 (Remote)

- Engineered and deployed a production CNN-based computer vision architecture utilizing ResNet and MobileNet configurations for automated defect detection on 10,000+ manufacturing image datasets.
- Boosted raw classification accuracy by 14% and curtailed inference latency by 35ms through comprehensive Hyperparameter Optimization (HPO), grid search scheduling, and customized data augmentation pipelines.
- Architected scalable training procedures using PyTorch DistributedDataParallel (DDP) and built automated ETL pipelines that cleaned, balanced, and pre-processed 50GB+ of multi-spectral image frames.
- Collaborated via structured Agile sprints using Git-driven version control to transition experimental models into highly documented deployment bundles.

Machine Learning Intern — CodeAlpha

1 May 2026 – 30 May 2026 (Remote)

- Designed and benchmarked two facial-emotion-recognition architectures on FER-2013 (35,000+ images, 7 classes) — a CNN built from scratch and a fine-tuned MobileNetV2 transfer-learning model — improving test accuracy from 52.27% to 59.46% via two-phase transfer learning.
- Corrected a 16.6× class imbalance through balanced class-weighting and targeted data augmentation, more than doubling minority-class F1-score on the hardest category.
- Optimized both models for edge deployment via TensorFlow Lite post-training quantization, achieving up to 108× inference speedup and a 74% reduction in model size.
- Built real-time inference applications — an OpenCV desktop app and a Flask web app with live webcam and photo-upload support — for end-to-end facial emotion detection.

PROJECTS

Aizen-GPT — GPT Built From Scratch

[GitHub Repo](#)

Python · PyTorch · Transformer Architecture · BPE Tokenization

- Implemented a GPT-style language model end-to-end — self-attention, multi-head attention, transformer blocks, and grouped-query attention — built on top of neural network primitives written from scratch (neurons, backprop, MLPs).

- Developed a custom BPE tokenizer, embeddings, positional encoding, and a KV-cache for faster inference; trained the model with a custom training loop and text generation script.

TechMart AI Support — Multi-Agent Customer Support Platform

[GitHub Repo](#)

Python · FastAPI · Next.js · LangChain · FAISS · Sentence-Transformers · SQLAlchemy

- Built a multi-agent customer support system routing messages to five specialized agents (Billing, Technical, Product, Complaint, FAQ) instead of one monolithic chatbot, with two-stage intent/sentiment detection (keyword classifier + LLM fallback).
- Grounded responses in a FAISS-indexed knowledge base via RAG; added JWT auth, persistent chat sessions, automatic support-ticket creation, and email/WhatsApp escalation notifications (SendGrid, Twilio).

Job Application Agent — Autonomous Job Search Automation

[GitHub Repo](#)

Python · LangGraph · LangChain · Ollama · Streamlit · GitHub Actions

- Built an autonomous agent that scrapes job listings (LinkedIn, Naukri, Internshala), scores them via semantic matching, identifies recruiter contacts, and orchestrates the workflow with LangGraph/LangChain using local LLMs (Ollama).

MediAgent — AI Medical Assistant

[GitHub Repo](#)

Python · FastAPI · Google Gemini API · React · Vite · Tailwind CSS · SQLite

- Built a full-stack AI medical assistant offering persistent chat, symptom triage, drug-interaction checks, appointment reminders, and medication adherence tracking.

EDUCATION

Bachelor of Technology — Computer Software Engineering

Aug 2023 – Jul 2027

Shankara Institute of Technology, Jaipur · *Relevant: Machine Learning, Deep Learning, Data Structures, Algorithms, DBMS*

CERTIFICATIONS

Machine Learning A-Z: AI, Python & R + ChatGPT Bonus — Udemy

Technology Job Simulation — Deloitte (Forage Virtual Experience Program)

ACHIEVEMENTS & ACTIVITIES

- Top 10 finish (out of 50+ teams) at a national-level hackathon for building MediAgent — AI Medical Assistant.
- Top 20 finish at a separate national-level hackathon, competing against teams from across the country.
- Solved 350+ data structures & algorithms problems on LeetCode, sharpening problem-solving speed and interview readiness.
- Active member of the Competitive Programming Club; regularly participate in online coding contests and ML competitions.
- Open-source contributor on GitHub — focused on LLM applications, RAG systems, and agentic AI workflows.